Week 9 Assignment

Repository Link:
https://github.com/Hughes-Tyler/AnalyticsCapstone

Report Link:
https://github.com/Hughes-Tyler/AnalyticsCapstone/blob/main/notebooks/main.ipynb

1. Background

Our goal is to predict the eventual status of a startup (operating, acquired, closed, or IPO) from its funding history, timing, location, and industry. The practical use is early screening: helping an investor or analyst flag which companies in a portfolio are most likely to reach a liquidity event or to fail, so that limited attention goes to the right places.

The data is a public startup dataset of 66,367 organizations. After restricting to companies founded between 1990 and 2016 (which removed legacy institutions such as universities and hospitals) and further cleaning, 49,638 companies remained. Each is described by 51 model inputs: 46 binary indicators (country, industry category, and two data-quality flags) and 5 principal components of the continuous funding and timing columns. The outcome is heavily imbalanced:

Status	Companies	Share
Operating	40,551	81.7%
Acquired	4,129	8.3%
Closed	3,971	8.0%
IPO	987	2.0%

Predicting "operating" for every company would be 81.7% accurate while finding no acquisitions, closures, or IPOs, so we judge models on macro F1 and per-class recall rather than accuracy alone.

In Weeks 6 to 8 we trained nine models: three Random Forests (shallow, baseline, deep), three Gradient Boosting models (shallow, moderate, deep), and three MLPs (V1 to V3). The problem was that each week scored its models with its own code, so the numbers weren't directly comparable. For Week 9 we reloaded all nine saved models and ran them through one

scoring function on the same validation rows (Section 3). Then we plotted training and validation log loss against model complexity to see which models underfit and which overfit (Section 4). We picked a winner using the validation set and ran only that model on the test set, once (Section 5). Section 6 covers what we took away from it.

2. Split Methodology

We made a single stratified 60/20/20 split (random seed 42) before any balancing, scaling, or modeling decision, so each set keeps the same class mix.

Status	Train	Validation	Test
Operating	24,330	8,110	8,111
Acquired	2,477	826	826
Closed	2,383	794	794
IPO	592	198	197
Total	**29,782**	**9,928**	**9,928**

Three rules protect the results from leakage:

1. **Split first.** Everything that learns from data was fit on the training set only: the scaler, the PCA, and the class weights.
2. **No SMOTE.** When we simulated SMOTE on the IPO training rows, 89.4% of the synthetic companies had at least one binary column that was neither 0 nor 1 (for example, half a country). Models that support class weights used balanced weights instead (IPO weight 12.58 versus 0.31 for operating), and the MLPs, which do not, were trained on oversampled copies of real minority-class companies. Every training row is therefore a real company.
3. **One-time test use.** Validation was used for every model comparison and for selecting the winner. The test set was touched once, by the winning model only, after the selection was locked.

3. Validation Evaluation of All Nine Models

The Week 6 Random Forests, Week 7 Gradient Boosting models, and Week 8 MLPs were each scored in their own weeks with different helper code. For Week 9 we reloaded all

nine saved models and ran them through one scoring function on the same validation rows. Nothing was retrained. Log loss is the chart metric because it reads predicted probabilities rather than hard labels, so it penalizes confident mistakes and makes the train-versus-validation gap visible.

Models are ordered by complexity rank, the number of learned parameters in the fitted object (tree nodes for the forests and boosting models, weights plus biases for the MLPs).

#	Model	Parameters	Train log loss	Val log loss	Gap	Val acc.	Val macro F1	AUC
1	MLP V1 - Small	900	0.993	1.059	0.066	0.55	0.369	0.78
2	MLP V2 - Moderate	5,540	0.645	1.624	0.979	0.61	0.375	0.71
3	RF V1 - Shallow	5,940	1.178	1.187	0.009	0.42	0.311	0.76
4	GB V1 - Shallow	5,944	1.024	1.049	0.024	0.53	0.365	0.80
5	MLP V3 - Large	54,724	0.399	2.768	2.369	0.67	0.388	0.69
6	GB V2 - Moderate	62,246	0.579	0.812	0.233	0.66	**0.435**	0.80
7	GB V3 - Deep	188,034	0.078	0.815	0.737	0.77	0.412	0.78
8	RF V2 - Baseline	586,056	0.700	0.830	0.130	0.67	0.425	0.80
9	RF V3 - Deep	4,072,746	0.117	**0.614**	0.497	0.81	0.318	0.80

Table 1. Validation results for all nine models, ordered simplest to most complex

Gap = validation log loss minus training log loss. A uniform 25% guess scores 1.386. AUC is one-vs-rest, macro-averaged.

Ranks run 1 to 9 with V1 < V2 < V3 inside each family, no model has a validation loss below its training loss, the Week 8 macro F1 values reproduce under the common path, and no test rows were used. Appendix A lists the remaining validation metrics.

4. Bias-Variance Analysis

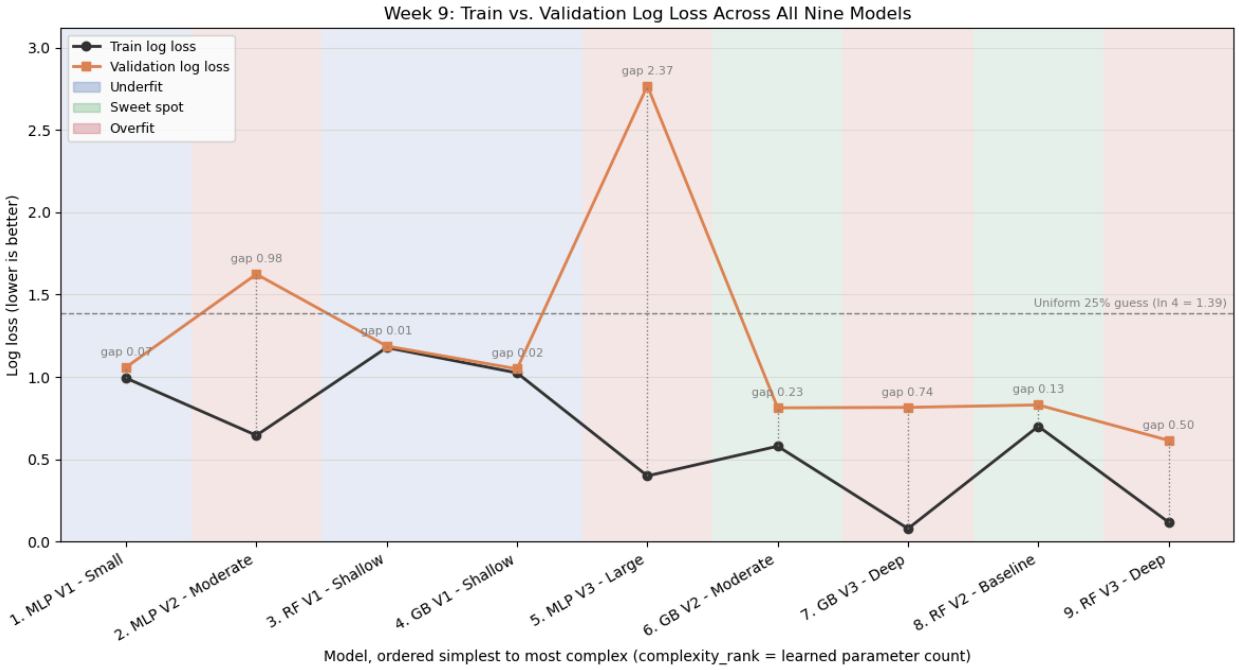


Figure 1. Train and validation log loss by model complexity, with underfit, sweet-spot, and overfit regions shaded

We labeled a model underfit if its training log loss was above 0.9 and its gap was under 0.1, overfit if the gap was over 0.4, and a sweet spot for everything in between.

The three V1 models (MLP V1, RF V1, GB V1) are underfit. Their training log loss is between 0.99 and 1.18 and their gaps are 0.07 or less, so they do about as badly on the rows they trained on as on new ones. They are just too simple to pick up the pattern. RF V1's validation log loss of 1.187 is not far from 1.386, which is what a uniform 25% guess scores.

MLP V2, MLP V3, GB V3, and RF V3 are overfit. GB V3 and RF V3 push training log loss almost to zero (0.078 and 0.117) while validation stays at 0.815 and 0.614, so the extra capacity mostly went into memorizing the training companies. The MLPs are worse. MLP V2 (1.624) and MLP V3 (2.768) both score worse on validation than a blind 25% guess would, so they are confidently wrong on companies they haven't seen.

Within each family the gap grows as the model gets bigger. Random Forest goes 0.009, 0.130, 0.497. Gradient Boosting goes 0.024, 0.233, 0.737. The MLP goes 0.066, 0.979, 2.369. Across all nine models the curve isn't a clean U, because parameter count is only a rough way to line up different model types. The MLPs start overfitting with far fewer parameters than the tree models, which is why they spike in the middle of the chart.

GB V2 and RF V2 land in the sweet spot. Both have validation log loss around 0.81 to 0.83 and moderate gaps (0.233 and 0.130), and they have the two highest validation macro F1 scores (0.435 and 0.425).

RF V3 has the lowest validation log loss at 0.614, but we don't treat it as the best model. Its validation accuracy of 0.81 is basically the 81.7% you get by predicting operating for every company, and its macro F1 is only 0.318. The class-balanced models push probability toward the small classes, and that costs them log loss on a validation set that is 82% operating. So we use log loss to show the gap and macro F1 to pick the winner.

Our read is that this task is mostly limited by what is in the features, and more rows probably wouldn't change much. Three very different model families level off in the same place (validation log loss around 0.81 to 0.83, macro F1 between 0.42 and 0.44), and extra capacity brings training loss down without helping validation. That is likely but not proven, since the overfit models would memorize the training data either way. The missing signal is probably in a few places: 59% of companies have only one funding round, the labels are a snapshot in time, the timing features may mostly be picking up company age, and the PCA may have thrown away some detail. Adding investor information or company-age controls would most likely help more than adding rows. Regularization would help the deep models, but it would mostly shrink the gap, and the ceiling would stay about where it is.

5. Winner Selection and Test Results

We selected Gradient Boosting V2 (moderate complexity) on validation, for four reasons:

1. **Best macro F1 of all nine models (0.435).** Macro F1 weights all four classes equally, which matches our goal of finding the rare outcomes. RF V3's lower log loss comes from favoring the majority class.

2. **Controlled gap (0.233).** GB V2 sits in the sweet-spot region. GB V3, the same model type with more capacity, drives training loss to 0.078, has a gap of 0.737, and scores a lower macro F1 (0.412). RF V2 has a smaller gap (0.130) and a very close macro F1 (0.425), so the two are near-equivalent; we prefer GB V2 for its higher F1, and we do not claim it has the smallest gap.
3. **The model family fits the data.** Gradient-boosted trees handle a mix of binary indicators and a few continuous components without extra scaling, are strong on tabular data, and respond well to class balancing for rare outcomes. The MLPs, by contrast, overfit severely and scored worse than a uniform guess.
4. **Strong ranking quality.** Its validation ROC-AUC (0.80) ties the best of any model.

These are interpretations supported by the evidence above. First, gradient boosting builds trees one after another, each correcting the errors of the last, which tends to extract weak signal from tabular data more efficiently than a neural network. Our inputs are mostly 0/1 indicators plus five components, a structure that trees split on naturally. Second, the moderate setting had enough capacity to learn real patterns without memorizing: GB V1 was too simple (macro F1 0.365), while GB V3 pushed training log loss to 0.078 and lost ground on validation (macro F1 0.412). Third, class balancing kept the rare classes influential during training, so the model learned to detect them instead of defaulting to operating (validation macro recall 0.54, against 0.30 for RF V3). Finally, its probability estimates were reasonably well behaved (validation log loss 0.812, ROC-AUC 0.80), unlike the MLPs, which performed pretty poorly on unseen companies.

Test-set results (touched once)

With the winner locked, we scored the test set once, with GB V2 only.

Split	Accuracy	Macro F1	Change in macro F1 from previous split
Train	0.7772	0.6676	n/a
Validation	0.6620	0.4345	-0.2331
Test	0.6610	0.4312	-0.0033

Table 2. Gradient Boosting V2 across train, validation, and test

Accuracy falls from 0.777 on training to 0.662 on validation, and macro F1 falls from 0.668 to 0.435. This large train-to-validation drop (0.233 in macro F1) is the overfitting already seen in the bias-variance analysis: the model fits the rows it trained on better than new companies. From validation to test the change is tiny: macro F1 moves from 0.4345 to 0.4312 (-0.003) and accuracy from 0.662 to 0.661. That second drop is the measure of selection optimism, the tendency of the best-of-nine model to look better on validation than it truly is, because we chose it for scoring highest there. It is negligible here. Macro precision (0.40) and macro recall (0.54) are also identical on validation and test, so validation performance was a fair estimate of how the model behaves on new companies.

Class	Precision	Recall	F1	Support
Operating	0.92	0.70	0.79	8,111
Acquired	0.27	0.52	0.35	826
Closed	0.22	0.47	0.30	794
IPO	0.20	0.46	0.28	197
Accuracy			0.66	9,928
Macro average	0.40	0.54	0.43	9,928
Weighted average	0.79	0.66	0.71	9,928

Table 3. Gradient Boosting V2 classification report on the test set

The model recovers about half of the rare outcomes on companies it has never seen: 52% of acquisitions, 47% of closures, and 46% of IPOs. The cost is precision. Only 20% to 27% of the companies it flags as acquired, closed, or IPO actually are, so it raises many false alarms. In the other direction, when it labels a company operating it is right 92% of the time, though it only catches 70% of operating companies because it deliberately leans toward flagging the rare classes. The normalized confusion matrix is in the notebook (Section 9.3).

The model identifies roughly half of acquisitions, closures, and IPOs on companies it has never seen, well above the naive all-operating rule, which finds none of them. That rule also scores 81.7% accuracy, which is why accuracy is the wrong yardstick; its macro F1 would be

about 0.22 against our 0.431. As a way to prioritize which companies deserve a closer look, the model adds value. It is not reliable enough to drive an investment or write-off decision on its own: with minority-class precision of 0.20 to 0.27, roughly four in five flagged companies will not have the predicted outcome, and its overall accuracy of 66.1% is below the naive baseline. It should be used as a ranked shortlist, not a verdict.

6. Summary

We compared nine models from three families using one scoring function and one fixed split. In every family, a bigger model brought training log loss down while validation log loss either stayed flat or got worse, and the gap between the two kept growing (Random Forest 0.009 to 0.497, Gradient Boosting 0.024 to 0.737, MLP 0.066 to 2.369). The two larger MLPs scored worse than a uniform guess. RF V3 had the best log loss, but mostly because it predicts “operating” for almost everything, so we went with Gradient Boosting V2 instead, since it had the best macro F1 (0.435) and did a better job on the rare classes.

On the test set, which we only used once, GB V2 got a macro F1 of 0.431 and accuracy of 66.1%. That is just 0.003 below validation, so picking the best of nine models on validation didn’t inflate the score much. I would use the model to build a shortlist of companies worth a closer look, but with minority-class precision of only 0.20 to 0.27 I wouldn’t act on its predictions alone. Three very different model families all leveled off at about the same validation score, which tells me the features are what’s holding us back, and a different algorithm or more rows probably wouldn’t move it much. If we kept going, the first things we would try are investor-level features, a control for company age, and a better way of representing companies with only one funding round.

Appendix

A. Additional validation metrics

Macro precision, macro recall, and macro F1 on the validation split for each model

#	Model	Val macro precision	Val macro recall	Val macro F1
1	MLP V1 - Small	0.37	0.55	0.369
2	MLP V2 - Moderate	0.36	0.46	0.375
3	RF V1 - Shallow	0.35	0.54	0.311
4	GB V1 - Shallow	0.37	0.57	0.365
5	MLP V3 - Large	0.37	0.45	0.388
6	GB V2 - Moderate	0.40	0.54	0.435
7	GB V3 - Deep	0.44	0.40	0.412
8	RF V2 - Baseline	0.40	0.50	0.425
9	RF V3 - Deep	0.53	0.30	0.318

B. Hyperparameters

- **Random Forest** (all with balanced class weights, seed 42): V1 Shallow uses 100 trees with maximum depth 5; V2 Baseline uses 200 trees with maximum depth 15; V3 Deep uses 300 trees with no depth limit.
- **MLP V3 Large:** hidden layers (256, 128, 64), alpha 1e-6, initial learning rate 0.01.
- The remaining Gradient Boosting and MLP settings are in notebook Sections 7.2 and 8.2.